

GaN Based Modified Integrated On-Board Charger Configuration Using Minimum Additional Active and Passive Components

Shohei Funatsu
Department of electrical
and mechanical
engineering
Nagoya Institute of
Technology
Aichi, Japan
s.funatsu.027@stn.nitech.a
c.jp

Hiroaki Matsumori
Department of electrical
and mechanical
engineering
Nagoya Institute of
Technology
Aichi, Japan
matsumori.hiroaki@nitech.
ac.jp

Takashi Kosaka
Department of electrical
and mechanical
engineering
Nagoya Institute of
Technology
Aichi, Japan
kosaka@nitech.ac.jp

Nobuyuki Matsui
Department of electrical
and mechanical
engineering
Nagoya Institute of
Technology
Aichi, Japan
nobu4357@gmail.com

Keisuke Nakamura
Aisin corporation
Aichi, Japan
keisuke.nakamura@aisin.c
o.jp

Mitsuru Takahashi
Aisin corporation
Aichi, Japan
mitsuru.takahashi@aisin.c
o.jp

Shin Taguchi
Aisin corporation
Aichi, Japan
shin.taguchi@aisin.co.jp

Subrata Saha
Aisin corporation
Aichi, Japan
subrata.saha@aisin.co.jp

Abstract— In this paper, a GaN based modified integrated on-board charger (IOBC) using a low value zero sequence inductance of traction motor and an inverter for class A and B segment electric vehicles (EV's) with dual power conversion stages is proposed. In the modified configuration, the traction motor and the inverter will be used as components in the back end DC to DC buck converter instead of in the front end AC to DC boost converter as available in the conventional IOBC configuration. The modified IOBC configuration satisfies the permissible THD distortion limit of the Japanese utility companies and also keep the output peak to peak battery current ripple within the specified limits. The feasibility of the modified IOBC configuration is verified from suitable simulation and experimental results.

Keywords—integrated on-board charger (IOBC), GaN, traction motor, E-axle

I. INTRODUCTION

Small passenger car production in 2022 for the class A and B segment consist about 15% in USA, 25% in Europe and 40% in Asia approximately [1]. There is a high possibility that by 2030 most of these class A and B segment vehicles in different countries will be converted to the electric vehicle (EV) version to satisfy the stringent CO2 emission standards [2]. Typical motor drive, battery and charger specifications of such class A and B segment EVs are given in Table 1 [3-5]. Slow residential charging at night from a single phase AC

supply via an integrated on-board charger (IOBC) placed within the E-axle [3] will be the most common way of charging the batteries of such EVs. There are two conventional concepts for the IOBC as shown in Fig. 1 and Fig. 2 respectively [5-8]. Both these concepts use bi-directional power converters with dual power conversion stages. Bi-directional power converters are preferred for supplying emergency residential load or feeding back energy to the grid. In the first concept, all the active and passive components used in the IOBC are new additional components and these components adds to the extra cost and volume of the E-axle. In the second concept, the same traction motor and the inverter are used as components in the front end AC to DC boost converter and hence, the extra cost and volume of the E-axle for the IOBC are greatly reduced. E-axes for the class A and B segment are highly cost and volume sensitive and

Table 1. Specifications of class A and B segment EV's.

Specification items	Range
Motor output	50~80 kW
Inverter Voltage	300~400 V
Battery capacity	20~30 kWh
Battery charger output	3.3~6.6kW

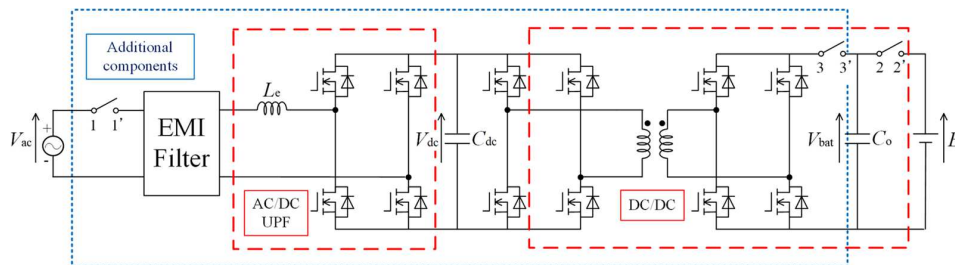


Fig.1 Conventional IOBC concept with dual active bridge.

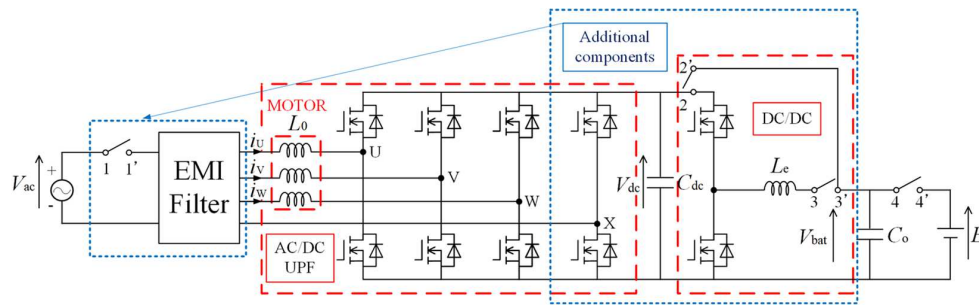


Fig.2 Conventional IOBC concept with traction motor.

hence, the second concept of an IOBC using the traction motor and the inverter is an attractive one. However, there are issues associated with the second concept of the conventional IOBC. Therefore, in this paper, first of all the main issue associated with the conventional IOBC configuration is highlighted and then a GaN based modified IOBC configuration is proposed.

II. IDENTIFICATION OF THE MAIN ISSUE

The main issue associated with the conventional IOBC configuration is the low zero sequence inductance of the interior permanent magnet synchronous motor (IPMSM) normally used as a traction motor. Due to the low zero-sequence inductance of the IPMSM traction motor, there is a generation of high total harmonic distortion (THD) in the grid current beyond the specified limits of the Japanese utility companies under low switching frequency and low control frequency operation of the front-end AC/DC boost converter.

Measurement of the zero sequence inductance of a typical IPMSM used as a traction motor for the class A and B segment electric vehicles is taken up to clearly identify the main issue. Specification of the IPMSM is given in Table 2. The H-Bridge circuit diagram used for the measurement of the zero sequence

inductance is shown in Fig. 3. The PWM voltage waveform with a 50% duty at 50 kHz switching frequency and the zero sequence current waveform (20A pp) across the zero sequence inductance of the IPMSM are shown in Fig. 4. The variation of the IPMSM zero sequence inductance with respect to switching frequency and the peak to peak current ripple is shown in Fig. 5. From the experiment result, it is clearly understood that the zero sequence inductance value of the IPMSM goes on decreasing with high switching frequency due to the presence of increased iron loss in the motor. Following the standard design procedures for a 3.6 kW battery charger available in several application notes [9], it is clearly understood that a minimum inductor value of 250 μ H coupled with a switching frequency and control frequency of 50 kHz is required for the front end AC/DC boost converter to satisfy the grid current THD limit specifications. The maximum switching frequency of the AC/DC boost converter is normally selected 50 kHz as the maximum control frequency of the IOBC controller considering a general purpose low cost micro-controller could not be increased beyond the 50 kHz value. The zero sequence inductance of the present IPMSM is

Table 2. Specification of the IPMSM.

Specification items	Range
Maximum torque	128.1 Nm
Maximum power	65.8 kW
Maximum speed	11000 r/min
Maximum current	132.1 Arms
Pole number	8
Slot	48

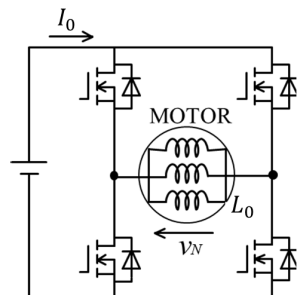


Fig. 3 H-Bridge circuit for the zero-sequence inductance measurement.

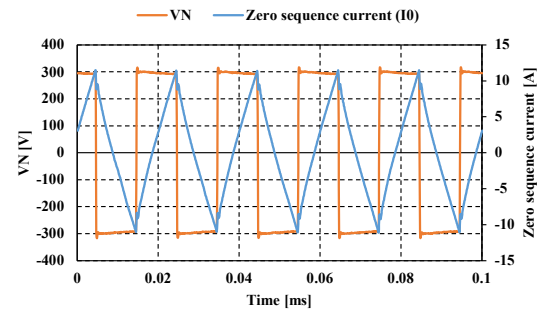


Fig.4 Zero sequence current waveform (20A_{pp}).

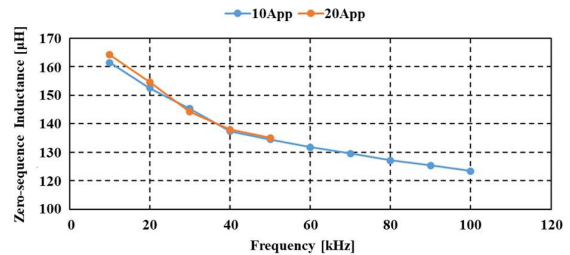


Fig.5 Zero sequence inductance values V.S. Switching Frequency.

135 μH at 50 kHz switching frequency and in future, for more high torque density IPMSM design the value of zero sequence inductance will become much lower. Since, the IPMSM traction motor could not satisfy the desired minimum zero sequence inductance value of 250 μH for a 3.6 kW battery charger, a GaN based modified IOBC configuration using the traction motor is proposed in this paper. Considering the high probability for the availability of low cost, high switching frequency GaN on Si HEMTS by 2025, they are the obvious choice as power devices for the IOBC and the traction inverter for class A and B segment EVs to satisfy the high power density target of the E-axle [10].

III. PROPOSED MODIFIED IOBC CONFIGURATION FOR LOW THD OF GRID CURRENT

The proposed modified IOBC configuration is shown in Fig. 6. In the modified IOBC configuration, the positions of the traction motor and the inverter are changed. In the modified IOBC configuration, instead of using the motor windings as an inductor in the front-end AC/DC boost converter, the motor windings are used as an inductor in the back-end DC/DC buck converter. Similarly, the external inductor (L_e) is used in the front-end AC/DC boost converter instead of the back-end DC/DC buck converter. In the modified IOBC, the value of the external inductor (L_e) could be suitably designed so that the harmonic distortion (THD) in the grid current is maintained within the specified limits of the utility companies for a specified switching frequency and control frequency of the AC/DC boost converter. On the hand, using the motor windings with low zero sequence inductance in the DC/DC buck converter side does not have any effect on

the output battery charging current ripple. The simple block diagram of the controller for the front-end AC/DC boost converter and the back-end DC/DC buck converter is shown in Fig. 7(a) and 7(b) respectively.

IV. SIMULATION RESULTS

The simulation of the conventional and modified IOBC configuration is carried out by solving the basic loop equations of the front end AC/DC boost converter and the back end DC/DC converter in the MATLAB environment. The parameter values of the conventional and modified IOBC configurations and the simulation conditions are given in Table 3. The simulation results of the grid current at 3.6 kW charging power output at battery load are shown in Fig. 8 (a) and 8 (b) for the conventional IOBC at 50 kHz and 100 kHz switching frequency respectively, where the control frequency of the IOBC controller is kept fixed at 50 kHz. The simulation result of the grid current at the same condition is shown in Fig. 8(c) for the modified IOBC at 50 kHz switching frequency and 50 kHz control frequency. The improvement in grid current THD distortion in the modified IOBC configuration compared to the conventional IOBC configuration is confirmed by the FFT analysis of the grid current waveforms in 8 (a-c). It is found that the THD (below 40th low order harmonic) of the grid current for the conventional IOBC at switching frequencies 50 kHz and 100 kHz is 6.9% and 7.1% respectively, which is above the 5% THD regulation limit of the Japanese utility companies. On the other hand, the THD (below 40th low order harmonic) of the grid current for the

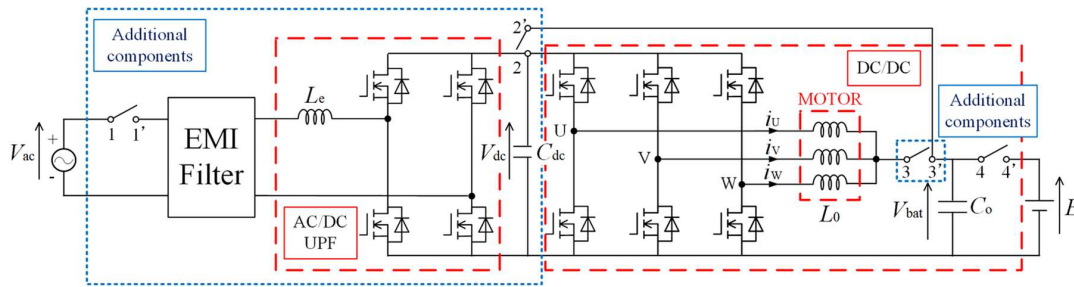


Fig.6 Proposed modified IOBC configuration.

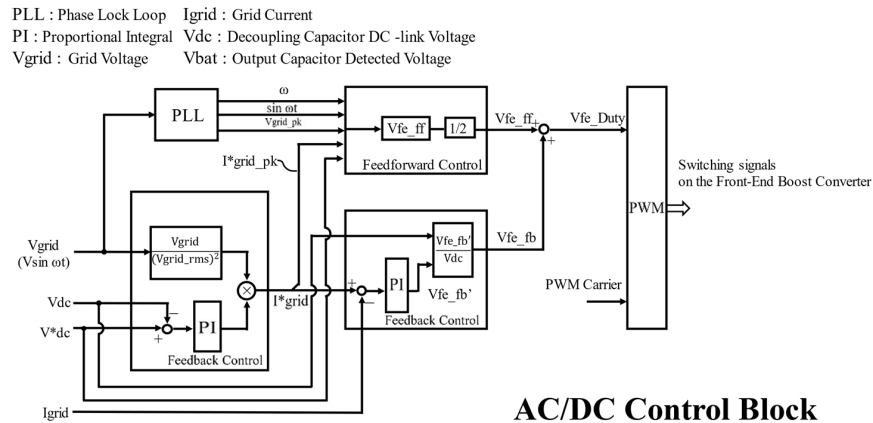


Fig.7(a) Block diagram of the controller for the front-end AC/DC boost converter.

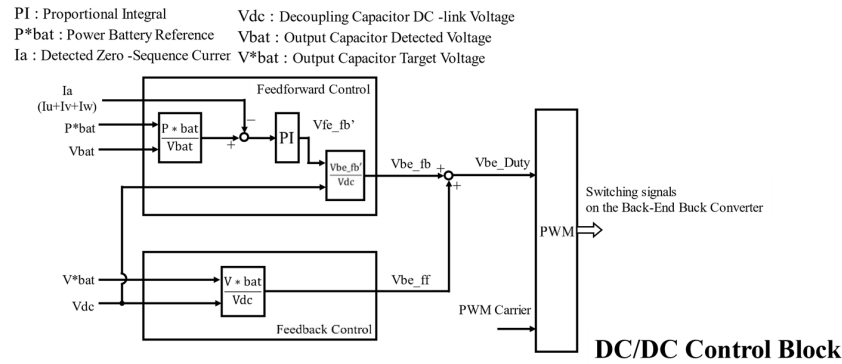


Fig.7(b) Block diagram of the controller for the back-end DC/DC buck converter.

modified IOBC at switching frequency 50 kHz is 2.9%. The improvement in the THD of the grid current for the modified IOBC configuration is mainly due to the 50% reduction of the lower order 3rd harmonic grid current. From the simulation results of the output battery current, it is also understood that the peak to peak battery current ripple current remains approximately the same for both the conventional and modified IOBC under the same above simulation condition of 50 kHz switching frequency and 50 kHz control frequency and is shown in Fig. 9 (a) and 9 (b) respectively. The obtained peak to peak battery current ripple of 1.2 A is within the specified limits of the battery manufacturers for safe battery operation.

V. EXPERIMENTAL RESULTS

For the preliminary experimental verification, discrete GaN devices (GaN System, Inc: GS66508T) in four parallel are used in both the conventional and IOBC configuration prototype. The IOBC including the traction inverter prototype is shown in Fig. 10. Grid simulator (NF: DP060RS) is used as a power source. The resistive load is used for output. The experimental results of the grid current at 3.6 kW (100% load) charging power output are shown in Fig. 11 and 12 for the conventional and the modified IOBC at 50 kHz switching frequency. In the modified IOBC, the grid current is well controlled with low THD (less than 5%). The zero current clamp in the grid current is due to the poor reverse conduction characteristics of the GaN devices [11]. The trends of the

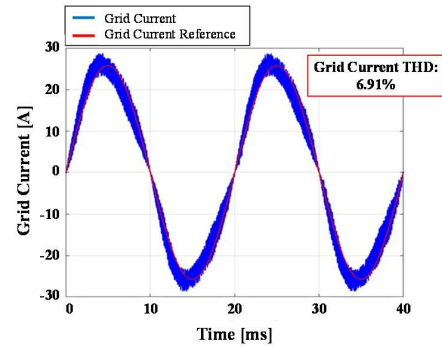


Fig.8(a) Simulation result of the grid current of conventional IOBC (3.6 kW, 50 kHz switching frequency and control frequency).

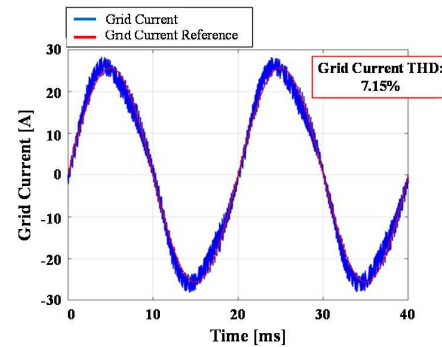


Fig.8(b) Simulation result of the grid current of conventional IOBC (3.6 kW, 100 kHz switching frequency, and 50 kHz control frequency).

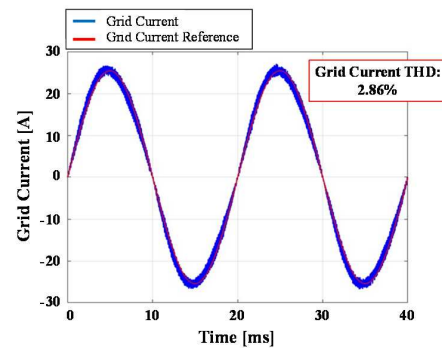


Fig.8(c) Simulation result of the grid current of modified IOBC (3.6 kW, 50 kHz switching frequency and control frequency).

Table 3. The parameter values and simulation conditions of the conventional and modified IOBC.

Main Simulation Parameters	Values
Motor zero sequence inductance (L_o)	135 μ H
External inductance (L_e)	250 μ H
DC link capacitance (C_{dc})	1000 μ F
Output capacitance (C_o)	500 μ F
Simulation conditions	Values
Power output	3.6 kW
AC Grid supply voltage	200 Vrms
DC link voltage reference (V_{dc})	340 V
Output voltage (V_{bat})	310 V
Zero sequence current reference	11.6 A

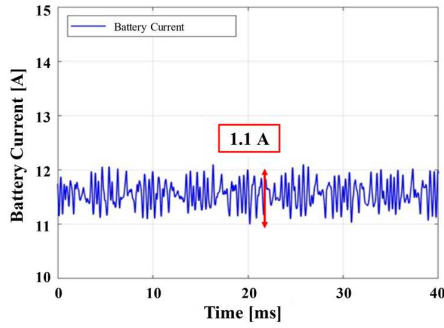


Fig.9(a) Simulation result of the peak to peak battery current ripple of conventional IOBC (3.6 kW, 50 kHz switching frequency and control frequency).

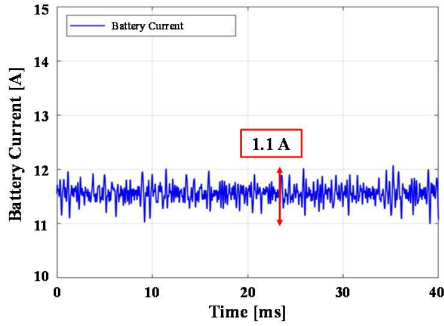


Fig.9(b) Simulation result of the peak to peak battery current ripple of modified IOBC (3.6 kW, 50 kHz switching frequency and control frequency).

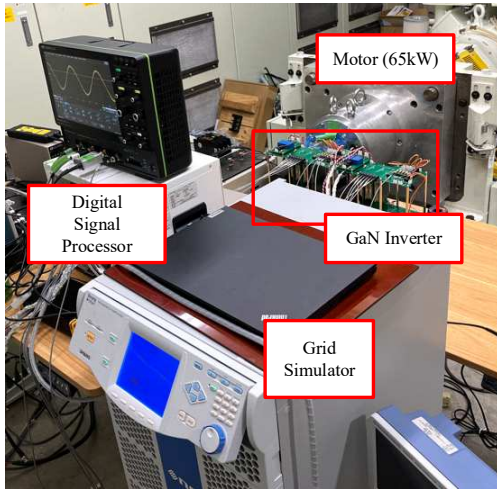


Fig.10 Overview of the experimental setup.

experimental results relating to the THD of the grid current for the modified IOBC match with the simulation result, and hence the merit of the modified IOBC compared to the conventional one is clearly understood. The experimental results relating to the THD of the grid current for different resistive load conditions for both IOBCs are shown in Fig. 13. Under conditions other than low load, the Modified IOBC is able to reduce the THD of the grid current more than the conventional OBC.

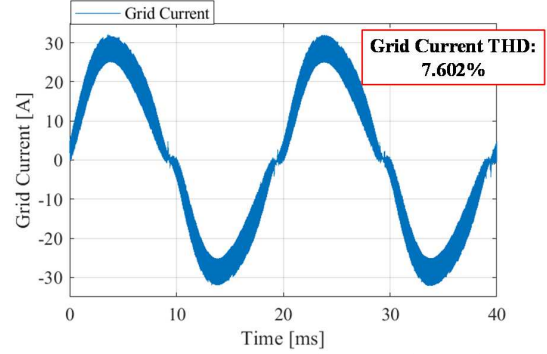


Fig.11 Experimental results of the grid current and the peak to peak battery current ripple of conventional IOBC (3.6kW).

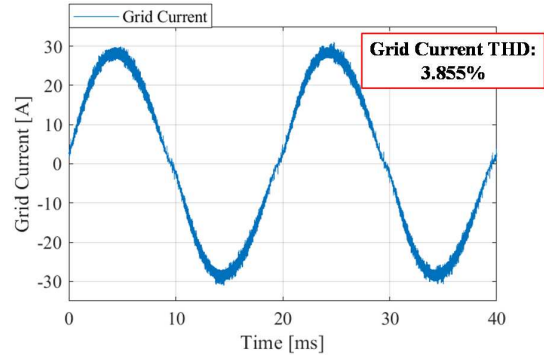


Fig.12 Experimental results of the grid current and the peak to peak load current ripple of modified IOBC (3.6kW).

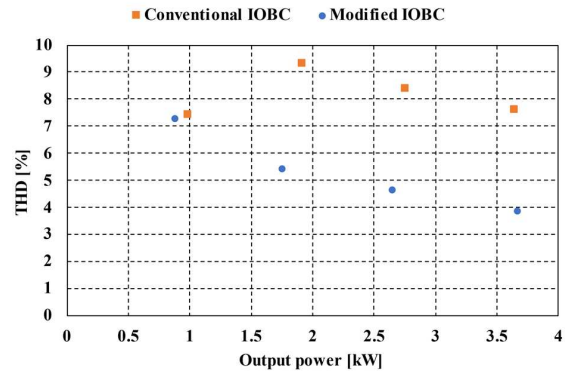


Fig. 13 Experimental results of THD of the grid current.

VI. CONCLUSIONS

The high grid current THD issue associated with the low zero sequence inductance of a typical IPMSM for the conventional IOBC is highlighted from suitable simulation and experimental results. Relevant simulation and experimental results are discussed showing that the above issues could be tackled with the proposed GaN based modified IOBC configuration.

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